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The Future of the SETI Post-Detection Protocols: Progress Towards Revisions

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Abstract

The International Academy of Astronautics has been the leading source of guiding principles to be followed in the event of a putative SETI signal detection. After many years of work, in 1989, the IAA formulated a set of Protocols, the “Declaration of Principles Concerning Activities Following the Detection of Extraterrestrial Intelligence”, which received widespread acceptance by the international scientific community. The IAA prepared a supplemental set of draft protocols in 1995 relating to the possible reply to a confirmed communication from an ETI, and both documents were presented in a position paper to the UN Committee on the Peaceful Uses of Outer Space.

The Search for Extra-Terrestrial Intelligence takes place in a dynamic context, and the Protocols are always a “work in progress” that must adapt to and reflect constantly changing scientific and social environments. Thus, the IAA SETI Protocols were streamlined and updated in 2010 after a multi-year process. Subsequently, the IAA SETI Committee has sought to review and revise the Protocols to keep pace with the expansion of search methods, the increased number of facilities and personnel conducting search activities worldwide, and the rapidly evolving information environment. In 2022, a Task Group was formed for this purpose, and a preliminary report was presented the following year during the 74th International Astronautical Congress in Baku. The relevance of this endeavour has been underscored in the past year by the deepening interest in what happens after the discovery of life beyond Earth, as evidenced by the 2024 workshops and symposia sponsored by the NASA Astrobiology Program, and the IAU/Kavli Foundation, the work of the SETI Post-Detection Hub at the University of St. Andrews, and the working group on “SETI and the Law” newly established by the International Institute of Space Law.

This report updates the activities of the Task Group following the 2023 Baku IAC, including further community consultations and proposed text, as we work towards a combination of timeless Principles reflecting the original intentions of the 1989 Protocols and supplementary Codes of Conduct and Best Practices documents that can provide additional guidelines that can be updated regularly to respond to changing circumstances.

Keywords: SETI, techno-signatures, Protocols, post-detection, discovery, Declaration of Principles

1. Introduction

The International Academy of Astronautics (IAA) Search for Extraterrestrial Intelligence (SETI) Committee was established in 1974 to bring together the international SETI community with the aims of exchanging ideas and updates about their science while also considering the global societal implications

of their work and related policy.[1], [2], [3], [4]. At the time, radio astronomy was the predominant search method. Early SETI proponents took seriously the responsibility they bore to consider the implications and ramifications raised by their astronomy research [5] while recognising that some questions and actions were beyond their authority and would require broad

interdisciplinary competence and international participation to resolve. The first Committee Chair, Rudolf Pešek, emphasised that it was essential “to have to have all important decisions concerning CETI¹ considered by the United Nations.” [6] From early on, there was substantial attention to the potential societal consequences of discovering extraterrestrial life² as well as to the issues surrounding active messaging to potential extraterrestrial recipients (METI) [8],[9],[10],[11],[12],[13],[14].

The main function of the IAA SETI Committee continues to be bringing together the SETI community, and since the 1990s, one of its main tasks has been to consider a post-detection framework with three distinct aims [15]: first, a set of guidelines to communicate a potential success of the experiment to public audiences; second, consideration of whether a reply should be sent to an incoming signal, and if so, what policies should be followed; and third, examination of the societal implications and longer-term consequences of a discovery. In 1989, the SETI Committee prepared the “Declaration of Principles Concerning Activities Following the Detection of Extraterrestrial Intelligence” (the Post-Detection Protocols) [16], which articulated guiding principles to be followed in the event of detection of a signal of putative extraterrestrial origin. This document was supplemented in 1995 by the “Draft Declaration of Principles Concerning Sending Communications with Extraterrestrial Intelligence (Reply Protocols)” [17],[15],[18],[19]. The Post-Detection Protocols were approved by the IAA Board of Trustees, and received broad acceptance and endorsement from the SETI community and international organisations including the International Institute of Space Law, the International Astronautical Federation, COSPAR, the International Astronomical Union, and Commission J of the Union Radio Scientifique Internationale.

In 1996, the IAA SETI Committee prepared the Position Paper, “Sending Communications to Extraterrestrial Civilisations” [20], which attached both the Post-Detection and the Draft Reply Protocols. The Position Paper was revised in 2007 [21]. The Paper “outlines an approach to an international process for deciding whether and how to send a response to an extraterrestrial civilization.” (p 2) The 1996 version of the Position Paper was approved by the IAA Board of

Trustees and the International Institute of Space Law. In 2000 it was presented to the UN Committee on the Peaceful Uses of Outer Space and referenced in the Report of COPUOS approved by the General Assembly [22]. This UN COPUOS Report noted that “while not necessarily requiring immediate action, [this issue] should be given serious consideration in connection with the future work of the Committee and its Legal Subcommittee” ([22]. pg 18, para 157). Nevertheless, the Draft Reply Protocols did not advance beyond the drafting stage.

The SETI Committee addressed this situation in 2010 with the preparation of the “Declaration of Principles Concerning the Conduct of the Search for Extraterrestrial Intelligence” (Declaration of Principles) [23], which updated the Post-Detection Protocols while avoiding specific details on the subject of sending a reply message. It quickly became apparent that the Declaration of Principles, while an update from the original Protocols, needed of further revision and updating to keep pace, *inter alia*, with the revolution in communications wrought by the Internet and social media. The SETI Committee initiated efforts to prepare an updated document in 2014, 2017, and again in 2019, which were unsuccessful for a variety of reasons.

In autumn 2022, the SETI Committee formed a new Task Group to update the 2010 protocols. A report on the activities of this Task Group was presented at the IAC in Baku in 2023 [24]. The work of the Task Group has continued, culminating in the preparation of a new Draft Revised Declaration of Principles Concerning the Conduct of the Search for Extraterrestrial Intelligence, which is presented by this paper.

2.1 Current Status of SETI Activities

The past decade has witnessed a resurgence in the search for extraterrestrial intelligence, propelled by the July 20, 2015, Breakthrough Foundation’s dedication of \$100 million to the Breakthrough Listen project. In addition, several state-of-the-art radio telescopes have entered into service in the past few years, including the Five-hundred-metre Aperture Spherical Telescope (FAST) in China, the MeerKAT array of sixty-four 13.5-metre diameter dishes in

¹ CETI is an acronym for Communication with ExtraTerrestrial Intelligence; this was the original term for this endeavour and for this Committee, later superseded by “SETI” or the Search for ExtraTerrestrial Intelligence.

² A few months after Frank Drake’s Project Ozma search in 1960, the NASA-commissioned Brookings Report [7] highlighted the need to study the potential societal consequences of discovering extraterrestrial life by radio searches.

South Africa, and the Canadian Hydrogen Intensity Mapping Experiment (CHIME) located in British Columbia. One year ago, the SETI Institute received a donation of \$200 million, which will assure that search efforts will continue for years to come.

This expansion of SETI research on an international scale has also driven an increase in conferences, symposia and other meetings for the exchange of information, many of which are interdisciplinary in nature. In the past few years, and especially this year, there have been several notable events on almost every continent, underscoring the global interest and participation in this endeavour.

The biannual *Astrobiology Science Conference*, sponsored by the NASA Astrobiology Program and the American Geophysical Union, brings together a wide range of scientists and some interdisciplinary researchers to focus upon the search for life beyond Earth, both within and beyond our own solar system, and research into life's origins. The 2022 meeting included sessions of specific relevance to SETI, e.g. on technosignatures (technologies and methodologies for radio and optical SETI), exoplanets, and "Scientists, Publics, Planetary Protection, and Post-Detection: Urgent Discussions for the 2020s". The 2024 meeting included, once again, sessions on technosignatures, exoplanets, and a large session on "Social Sciences, Humanities, and Astrobiology" with diverse papers including some concerning public communication and potential reactions to discoveries of extraterrestrial life. AbSciCon sessions and plenaries in 2022 and 2024 addressed standards of evidence for biosignatures, and communication of astrobiology results to broad audiences -- entangled subjects receiving sustained attention from senior astrobiology scientists and administrators.

Of particular significance, in March 2024, the NASA Astrobiology Program sponsored a large virtual workshop, *Communicating Discoveries in the Search for Life in the Universe* [25], with participants with expertise in astrobiology science (including SETI), science communication practice and research, and social research, ranging from Early Career Researchers (ECRs) to senior scholars. Workshop participants spoke in detail about prior putative discoveries of extraterrestrial life and the challenges for scientists in these situations, as well as looking at potential future discovery scenarios. We have taken note of this workshop in our current Protocols draft and look forward to the collectively authored workshop report forthcoming in 2025, which is

expected to include concrete recommendations for individuals, agencies, and institutions responsible for communicating information about the search for and detection of life beyond Earth.

Continuing on from the Communicating Discoveries workshop was the "Breaking News: We Found Extraterrestrial Life!" week-long workshop at the Lorenz Centre in Leiden in Sept 2024, led by researchers from Leiden University, the NASA Astrobiology Program, and the Earth-Life Science Institute in Tokyo.[26] This work is expected to continue in the years ahead.

Another important trend is the gradual reintegration of technosignatures research into NASA astrobiology work, starting with the NASA Technosignatures Workshop in 2018 [27], continuing through additional workshops, and the ongoing Technosignature Seminar Series at NASA Goddard [28]. Sessions at the Penn State SETI Symposia in 2022 and 2023 specifically addressed the potential for future NASA missions to include technosignatures work. An upcoming October 2024 meeting hosted by the NASA Astrobiology Program, *Astrobiology and the Future of Life* [29] boasts a program including astrobiology, multiple papers on technosignatures science, and papers on cultural dimensions and social research related to astrobiology and SETI.

The Penn State SETI Symposium in 2022 and 2023 included many papers on technosignatures science, a few on interdisciplinary subjects, and invited talks and extended breakout sessions on post-detection. 2022's breakout session on "Post-detection for the 2020s: fundamentals, current concerns, and new discussions" was followed in 2023 by "The Alien in the Machine: Post-Detection in the Era of AI". Each session had a large team of co-organisers, including IAA SETI Committee members and members of the SETI Post-Detection Hub (described below).

The Kavli Foundation and the International Astronomical Union (IAU) co-sponsored the IAU Symposium (IAUS 387), *(Toward) Discovery of Life Beyond Earth and its Impact*, which was held in the UK in April at Durham University. The Symposium rationale stated that

A new era of exploration has begun, rich in data derived from these new technologies. We can search the universe, looking for life and, in the process, learning more about our ourselves, our planet, our species, and perhaps our significance.

... We are not prepared for a discovery. We need to bring together diverse expertise to plan how we will assess evidence and communicate what we know (and don't know) with the public. [30]

The Symposium brought together experts from diverse disciplines crossing boundaries of anthropology, astrobiology, astronomy, chemistry, ethics, law, media communications, philosophy, policy, and theology, for an in-depth examination over five days. Although the theme was the discovery of evidence of life broadly, there was a special focus on SETI and the discovery of intelligent life. Several members of the IAA SETI Committee were co-organisers of the Symposium, presenters, or both, including three of the present authors: Carol Oliver was an organiser and conducted a mini-workshop and spoke about how we can effectively communicate our conclusions. Kathryn Denning discussed “The problems of preparation”, focusing upon the utility of social research, safety for scientists, and the need to build professional capacity to meet post-detection challenges. Leslie Tennen presented on International Space Law and SETI.

Among the thought-provoking concepts discussed at the Symposium were questions such as: What is life? What is consciousness? And what is intelligence? At what point does a collection of cells become a living entity? At what point does the chemistry of life begin? What about non-carbon-based entities? And what about artificial life, i.e., machines? Can machines be considered as a life form? Is it possible for machines to achieve consciousness? Do the rules of metalaw apply to artificial life? Our definition of “life” has a significant impact on the methods of the search and the possible evidence that may be discovered. The Symposium easily could have gone beyond the scheduled five days of presentations, discussions, and workshops. Fortunately, the Symposium proceedings will be published by Cambridge University Press, with an anticipated release by early 2025.

The European Astrobiology Network Association (EANA) 2024 Conference was held from September 3 - 6 in Graz, Austria. Sponsored by the European Space Agency (ESA) in conjunction with European research institutions, the meeting was designed to bring together the solar system and the exoplanet communities to discuss planetary protection issues and the origins of life, both on Earth and potentially elsewhere.

Other meetings include the ongoing activities of the SETI Institute including Earth at the Crossroads II (April 2024), the annual International Astrobiology Workshop, the Life in the Universe conference in Capetown (August 2024), and of course the IAA Symposium on SETI held during the International Astronautical Congress each year.

The Breakthrough Foundation continues to support the highly successful Breakthrough Discuss conferences, with the latest meeting in Oxford, UK in July 2024. These large, multidisciplinary efforts focus on exploring life in the Universe and developing innovative ideas for space exploration. In 2023, the Breakthrough Listen Initiative [31] also relocated its headquarters to Oxford University [32], with funding secured for an additional six years.

Educational institutions around the world have initiated programmes in the discipline of astrobiology. Examples include the Australian Centre for Astrobiology at the University of New South Wales in Australia, the Arizona Astrobiology Center (AABC) at the University of Arizona, the University of Edinburgh, the University of Tokyo, the Pierre and Marie Curie University in France, the Centro de Astrobiología (CAB) in Spain, Arizona State University, the University of Washington, the University of California at Berkeley, the University of Chicago, and the Nordic Network of Astrobiology Graduate Schools in Scandinavia. The subject of SETI or technosignatures searches is frequently addressed in generalist “Life in the Universe” university courses and is taught at the graduate level by astronomers at a handful of institutions, including Penn State, and via internships with the SETI Institute and Breakthrough Listen/ University of California Berkeley.

2.2 Groups working on human factors

Several groups are strongly engaged in the human factor elements of post-detection—the public reactions to the putative discovery after its initial communication.

In fall 2022, the SETI Post-Detection Hub was launched by two members of the IAA SETI Committee, John Elliott and Martin Dominik at the University of St. Andrews in Scotland, with members of the UK SETI Research Network, the IAA SETI Committee, and additional international members and partner institutions [33]. As of Sept 2024, partners in this Hub are the St. Andrews Centre for Exoplanet Science and the Centre for Global Law and

Governance, the Durham University Law School, and the Jodrell Bank Centre for Astrophysics of the University of Manchester. Currently, eleven members of the IAA SETI Committee are involved with the SETI Post-Detection Hub, including all four of the authors of this paper. The following statement on the website summarises the purpose of the Hub:

The time is thus right for consideration of humanity's response – and responsibility – following the detection of both life and intelligence in the Cosmos. We should plan now for this eventuality by setting out impact assessments, protocols, procedures and treaties designed to allow humanity to respond responsibly. [34]

More recently, the International Institute of Space Law established a Working Group on SETI and Law. This group launched in January 2024 under initial convenor Michael Bohlander, but as of September 2024, it will go forward under the leadership of Andrea Harrington. The Working Group is scheduled to run until 2026. The *raison d'être* for the Working Group is to:

...contribute to laying the legal groundwork for providing guidance to relevant domestic and international stakeholders in government and civil society, regarding the analysis and development of regulatory protocols [35]

Details of the Working Group's focus and outputs are evolving and will be available in future years, but generally, it is focused upon legal issues that would follow a confirmed detection of extraterrestrial intelligence. Several IAA SETI Committee members are involved in the Working Group, including present authors Michael Garrett, Leslie Tennen, and Kathryn Denning, and John Elliott and Jason Wright.

2.3 Building Capacity for Post-Detection Efforts

The Protocols revision process is an opportunity for reflection. As noted above, there are many efforts underway circling around the idea of improving our collective capacity for post-detection work of different kinds. Decades of work in the scientific searches for life beyond Earth have resulted in a very complex field for scientists, interdisciplinary researchers, legal

scholars, and policy experts alike, but for now, post-detection-related projects remain disparate, ephemeral, and driven forward primarily by individuals rather than organisations [36].

Interdisciplinary consultations clearly are required both for issues of policy as well as for the societal implications and impacts of potential discoveries of extraterrestrial life. There has now been over 60 years of policy work related to SETI and METI, both within and outside of the IAA SETI Committee. Consequently, there is a great deal of history; however, there is no centralised repository or database of this work. There has been neither a systematic effort at integrating work from different countries, nor much attention to translation to or from English. There is a compelling need to consolidate this reference material, summarise it, and maintain it in accessible archives for Committee members and others to draw upon.

The probability of achieving these goals would be greatly enhanced by support in three forms: funding for travel to meetings; administrative support for organizing meetings and preparing proceedings and other reports; and research assistance. SETI has entered a new era, where scientific experiments are on reasonably secure financial footing for the next several years. Policy and societal issues also require renewed attention and financial support. Nevertheless, we must remain mindful that while support from the organisations conducting or otherwise interested in SETI is needed, so is some independence from them.³

3. Draft Revisions to the 2010 Declaration

The discussion above is by no means an exhaustive recitation of all the recent astrobiology and SETI activities, but is representative of the burgeoning global interest in the subjects of potential societal impacts, sociopolitical issues, legal issues, and questions about optimal ways for scientists to communicate with the public. It is apparent that the scope of the current activities of these entities and meetings demonstrate a growing need for the leadership of the IAA SETI Committee, and that appropriately revised and updated Protocols are both relevant and essential. Although the scope of activities by these other entities and meetings overlaps with the IAA SETI Committee, the IAA SETI Committee is

³ Allan E. Goodman, one of the architects of the first Protocols, first made this observation and suggestion in 1990 in relation to a standing post-detection commission: "These administrative costs should be financed by an endowment raised from donations by governments and private persons and foundations" [37].

unique in its exclusive, long-term focus on potential future technosignature discoveries.

The present endeavour to update the Protocols builds on the work of previous efforts, which sought broad participation that required administrative and financial support which, unfortunately, could not be obtained. Nevertheless, the Task Group continued to move forward. Led by Carol Oliver and Kathryn Denning, the members of the Task Group are the IAA SETI Committee officers in addition to Carol Oliver (Co-Chair), that is Michael Garrett (Chair), Andrew Siemion (Co-Chair), and Lori Walton (Secretary), and Leslie Tennen (IAA Legal Counsel). Pertinent literature has been reviewed to glean the benefits of the extensive body of work that has been devoted over the years to post-detection policies and issues [36],[38],[39],[40],[41],[42],[43],[44]. Comments, suggestions and concerns have been discussed across interdisciplinary lines with scientists, especially astronomers, social scientists and media experts.

These activities have culminated in the preparation of a Draft Revised Post-Detection Declaration of Principles, which is appended to this paper and formally presented to the IAA SETI Committee and the international scientific community. This Draft is respectfully submitted for discussion and comment, with the intention to have a finalised document to present to the IAA SETI Committee for consideration and approval at the 76th International Astronautical Congress in Sydney in October 2025, whereupon it will be submitted to the IAA Board of Trustees for endorsement. The method and process for the discussion and comments are to be determined by the IAA SETI Committee executives.

The Draft Revised Declaration updates the current 2010 version, and addresses several matters that were not previously mentioned, including clarity in identifying the parties to the document, engagement with social media, and an express reference to technosignatures. The Draft Revised Declaration, in its present form, commits the IAA SETI Committee to establishing a qualified Post-Detection Sub-Committee, while reserving the details for future discussion. It does not yet address all the issues identified by community members and described in our previous 2023 IAC paper [24], as these are myriad

and complex. Further, the document defers the issues regarding a global response to the detection of confirmed extraterrestrial communication, and does not address the separate and distinct subject of messaging to extraterrestrial intelligence in advance of a confirmed detected extraterrestrial signal (METI).

Whether or not to send a global response to a confirmed detection of an extraterrestrial signal, and the content of any such response, and whether or not messages should be sent in advance of a confirmed detection, are controversial and contentious issues, which have been the subject of debate for decades.⁴ [45],[46],[47],[48],[49],[50],[51],[52].

The IAA SETI Symposium for this 75th IAC in 2024 has multiple papers scheduled which address messaging [53],[54],[55],[56],[57], which underscores the need for concentrated and coordinated effort from the SETI Committee in addition to these disparate and standalone efforts.

As there is entrenched disagreement on these matters within the SETI community, the inclusion of specific directives one way or another could present an insurmountable obstacle to achieving wide acceptance of the document as a whole. Rather than delay progress on updating and revising the Protocols, the Draft Revised Declaration simply provides that signatories shall refrain from unilateral communications with a confirmed ETI and cooperate with appropriate international consultations conducted through a broadly representative international body, such as the United Nations, with specific procedures to be outlined in a separate agreement, declaration, or arrangement to ensure a coordinated and responsible approach.

The language in the Draft Revised Declaration is a compromise in certain places, in order to facilitate the release of the document for consideration and robust discussion, and should not be interpreted as a blanket endorsement by any of the authors.

With this presentation of the Draft Revised Declaration, the work of the Task Group is completed. We thank all those who participated in the efforts to update the Protocols.

⁴ While advocating international consultations and a moratorium on METI, Billingham and Benford noted in 2011 that “No one seems opposed to having international consultations about transmitting *after* we detect them by standard SETI. Assuming this to be the case, it is surely even more important to have the consultations about transmitting *before* we detect them when we don't even have their signal in hand.” [45]

DRAFT REVISED Declaration of Principles Concerning the Conduct of the Search for ExtraTerrestrial Intelligence (SETI)

The parties to this Declaration are individuals and institutions participating in the scientific Search for Extraterrestrial Intelligence (SETI), that is, the astronomy-based search for ‘technosignatures’¹ or evidence of past or present intelligent life and technology beyond Earth.

The purpose of this Declaration is to declare our commitment to conduct the Search for ExtraTerrestrial Intelligence in a scientifically rigorous manner; to establish uniform procedures for scientists to confirm putative evidence of an intelligent extraterrestrial civilisation; to provide guidance to the scientific community for the announcement of a confirmed SETI detection while balancing the community imperative of providing rapid and accurate information to a wide audience, with consideration for the safety and exposure of individual scientists involved; and to proactively inform the public of the procedures and guidelines that signatories will respect if a confirmed SETI detection occurs.

The commitments in this Declaration are made with the recognition that the scientific and communications landscapes are ever-changing, and this Declaration will be supplemented with Best Practices Guidelines and a Code of Conduct that will be periodically re-examined and updated. This Declaration and any supplemental Guidelines and Codes, together with a current list of signatory parties, will be placed on file with the International Academy of Astronautics (IAA) and made available on the IAA website.

Principles

1. Handling Candidate Evidence:

- In the event of a putative detection of extraterrestrial intelligence, the discoverer will make all efforts to verify the detection, using the resources available to the discoverer and in collaboration with other investigators. Such efforts will include, but not be limited to, observations at more than one facility and/or by more than one organisation.
- Information about candidate signals or other potential detections should be handled with the same care as provisional laboratory results, recognising that initial findings may be incomplete or ambiguous, requiring thorough analysis and confirmation, which could be a lengthy process. It is crucial to uphold the highest standards of scientific responsibility and integrity throughout this process. Additionally, best practices in science communication should be employed to clearly convey the importance and significance of candidate discoveries to non-specialist audiences.

2. Searching:

- SETI practitioners and their institutions and organisations will be free to present reports on activities and results in public and professional fora. Institutions and organisations should be responsive to news organisations, social media platforms, and other public communications media. Responses to inquiries should be prompt, accurate and honest.
- There is no obligation to disclose verification efforts until a discovery is confirmed. If a candidate technosignature is discovered, communication about ongoing observations and analyses may be necessary to dispel rumours and provide accurate and reliable information.

¹ ‘Technosignatures’ are defined as observable evidence of technology utilised by advanced extraterrestrial civilisations e.g. narrow-band radio signals, laser emission, infrared excess associated with large-scale energy usage, anomalies in astronomical measurements due to megastructures etc. Unlike biosignatures, which point to biological processes, technosignatures would indicate the presence of intelligent life and focus on artificial or engineered phenomena that would directly reveal the existence of extraterrestrial civilisations. This Declaration is intended to apply to the search for intelligent extraterrestrial life, not extraterrestrial life in general, nor to unidentified anomalous phenomena (UAP) in the Earth’s atmosphere.

- Speculative or unconfirmed conclusions should be clearly identified as such, and exaggerated statements should be avoided.
- Institutions and organisations should engage with news organisations to provide accurate and timely updates of their activities, ensuring proactive engagement with the public to provide a source for accurate information and consider other ways to explain a putative signal among public audiences.
- Individual practitioners should have the right to decline from engaging directly or continually with the media, including social media if they choose to do so, and shall use their best efforts to ensure that their organisation or institution provides updates on their science.

3. Confirmed Detections:

- If the verification process confirms – by the consensus of the other investigators involved and to a degree of certainty judged by the discoverers to be credible – that a signal or other evidence is due to extraterrestrial intelligence, the discoverer shall report this conclusion in a full, complete and open manner to the public, the scientific community, and the Secretary General of the United Nations.
- The confirmation report should include the basic data, the data analysis process and results of the verification efforts, any conclusions and interpretations, and any detected information content of the signal itself. A formal report will also be made to the International Academy of Astronautics, the International Astronomical Union Central Bureau for Astronomical Telegrams, the United Nations Secretary General, the Committee on Space Research (COSPAR) of the International Science Council, the International Institute of Space Law, and the International Telecommunications Union.

4. Data Accessibility:

- All data necessary for the confirmation of the detection to be replicated should be made available to the international scientific community through publications, meetings, conferences, official websites, and other appropriate means.

5. Monitoring and Archiving:

- The discovery should be monitored and best practices for the safe, reliable and resilient handling of data should be employed. Any and all data bearing on the evidence of extraterrestrial intelligence should be recorded and stored permanently to the greatest extent feasible and practicable, in a form that will make it available to observers and to the scientific community for replication of results and further analysis.

6. Data and Frequency Protection:

- If the evidence of detection is in the form of electromagnetic signals, observers should seek international agreement to protect the appropriate frequencies by exercising the extraordinary procedures established within the International Telecommunication Union. Evidence in a form other than electromagnetic signals should be protected utilising best scientific practices.

7. Post-Detection Protocol:

- The IAA SETI Committee will appoint a Post-Detection Sub-Committee to assist in matters that may arise in the event of a confirmed signal, and to support the scientific and public analysis by offering guidance, interpretation, and discussion of the wider implications of the detection. This sub-committee will be chaired by the chair of the IAA SETI Committee, and should include communications professionals.
- The SETI Committee Executive Officers will support and assist the Post-Detection Sub-Committee with engagements with social media platforms and news organisations to responsively and effectively assist in the dissemination of accurate and reliable information.
- The IAA SETI Committee will maintain and support working groups dedicated to Post-Detection issues, providing assistance during the discovery process, aiding in scientific analysis, and focusing on subjects such as best practices for public communication about technosignatures science. These working groups will actively collaborate with interdisciplinary researchers to explore the broader implications of potential discoveries.

8. Communications with ETI following a confirmed detection:

- Signatories to this Declaration will refrain from making any unilateral communications in response to a confirmed extraterrestrial signal, and shall cooperate with appropriate international consultations. These consultations should be conducted through a broadly representative international body, such as the United Nations. The specific procedures for such consultations will be outlined in a separate agreement, declaration, or arrangement to ensure a coordinated and responsible approach.²

9. Ethical and Legal Considerations:

- SETI practitioners shall adhere to the highest ethical standards, ensuring cooperation, honesty, and integrity in all aspects of their work. They will collaborate with international legal authorities to establish clear frameworks for managing the dissemination of information about potential extraterrestrial detections, ensuring compliance with relevant laws and regulations. SETI practitioners shall cooperate in establishing ethical standards for the handling of any putative and confirmed detected evidence of extraterrestrial intelligence.

This Declaration shall be open for signature by organisations and practitioners engaged or otherwise interested in SETI. These revised Protocols are intended to replace the previous document adopted by the International Academy of Astronautics SETI Committee in 2010.

² This Declaration is not addressed to the separate and distinct subject of messaging to extraterrestrial intelligence in advance of a confirmed detected extraterrestrial signal (METI).

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